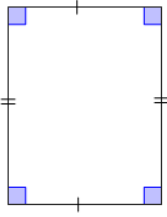


Classifying quadrilaterals

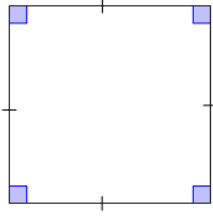


- 1 Give the most specific classification of the following quadrilaterals as trapezoid, rectangle, or square:

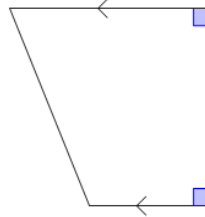
a



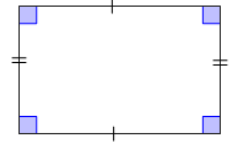
b



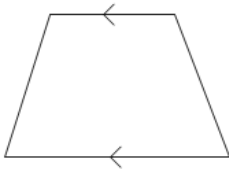
c



d

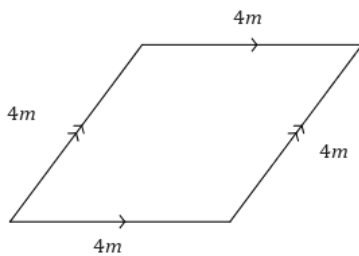


e

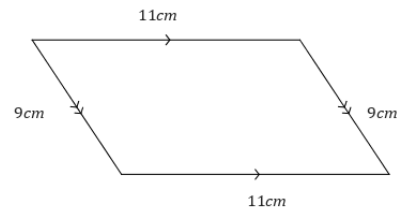


- 2 Give the most specific classification of the following quadrilaterals as parallelogram, trapezoid, or rhombus:

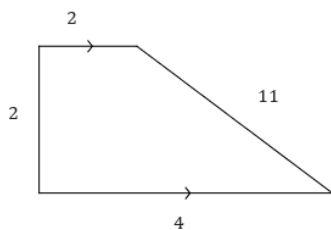
a



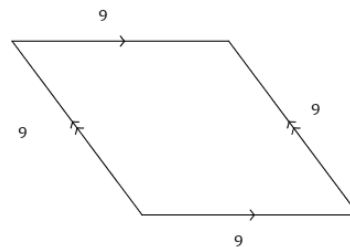
b



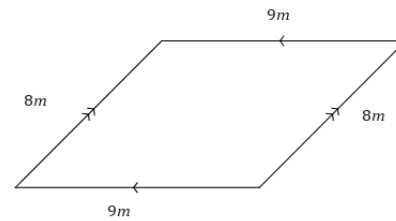
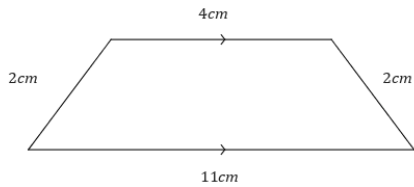
c



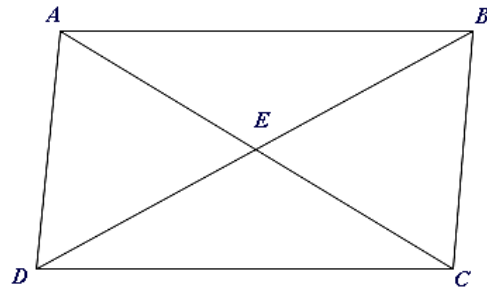
d



e



- 3 In the given figure, $\triangle AEB \cong \triangle CED$.
What type of quadrilateral is $ABCD$?



- 4 Determine whether each statement is true or false.
- a A trapezoid is a parallelogram.
 - b A square is a rhombus.
 - c A parallelogram is a rectangle.
 - d A kite is a rhombus.
 - e A rhombus is a square.
 - f All parallelograms are quadrilaterals.
 - g All quadrilaterals are parallelograms.
 - h All rhombi have a pair of parallel sides.
 - i All quadrilaterals with a pair of parallel sides are rhombi.
 - j All squares are parallelograms.
 - k All rhombi have congruent consecutive sides.
 - l All parallelograms are squares.
 - m All quadrilaterals with a a pair of congruent consecutive sides are rectangles.



5 For each of the following scenarios, state the most specific classification of the quadrilateral.

- a** Patricia draws a quadrilateral, and covers it up. She tells Glen that the quadrilateral consists of right angles only.
- b** Hannah draws a parallelogram, and covers it up. She tells Glen that the opposite sides are parallel.
- c** Aaron draws a parallelogram, and covers it up. He tells Kathleen that consecutive sides are congruent and the diagonals are perpendicular.
- d** Marge draws a parallelogram, and covers it up. She tells Peter that the figure has all right angles.
- e** Marge draws a parallelogram, and covers it up. She tells Paul that the diagonals of the parallelogram are perpendicular.
- f** Bob draws a parallelogram, and covers it up. He tells Emma that the diagonals bisect each other.
- g** Tom draws a parallelogram, and covers it up. He tells Rosey that consecutive sides are congruent.
- h** Michael draws a parallelogram, and covers it up. He tells Xanthe that consecutive angles are supplementary.

6 For each of the following descriptions, state the most specific classification of the quadrilateral.

- a** A quadrilateral with opposite sides that are congruent.
- b** A quadrilateral with all angles congruent.

7 For each of the following descriptions, give an additional information to guarantee that the quadrilateral is a parallelogram.

- a** A quadrilateral with one pair of opposite sides that are parallel.
- b** A quadrilateral with perpendicular diagonals.
- c** A quadrilateral with one diagonal bisects the other.

8 Determine what must be shown to prove that the parallelogram is:

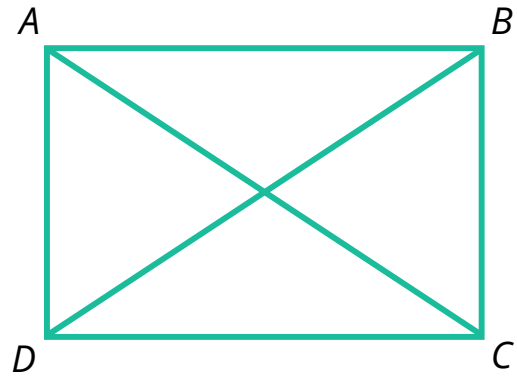
- a** Rhombus
- b** Rectangle
- c** Square

9 Determine what must be shown to prove that a given trapezoid is isosceles in terms of:

- a** Diagonals
- b** Base angles
- c** Opposite angles

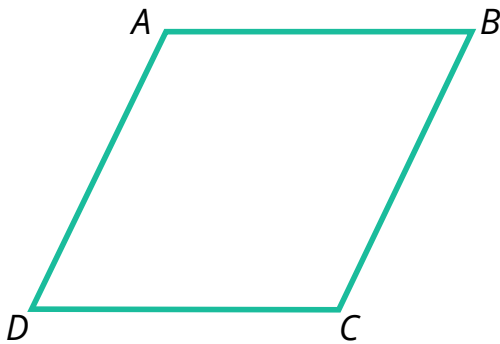


- 11 Given that $ABCD$ is a parallelogram, and $\overline{AC} \cong \overline{BD}$, prove that $ABCD$ is a rectangle.

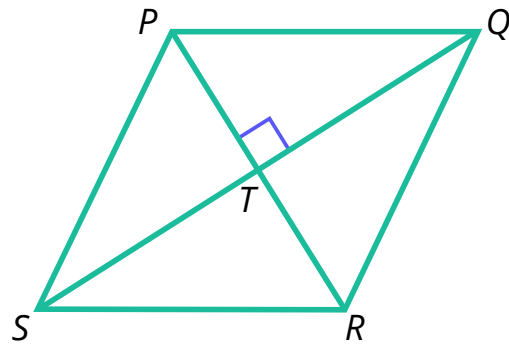


- 12 Prove that the following is a rhombus with the given conditions:

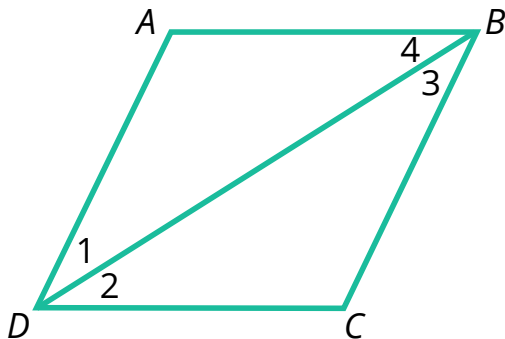
- a Given that $ABCD$ is a parallelogram, and $\overline{AB} \cong \overline{BC}$.



- b Given that $PQRS$ is a parallelogram, and $\overline{PR} \perp \overline{QS}$.



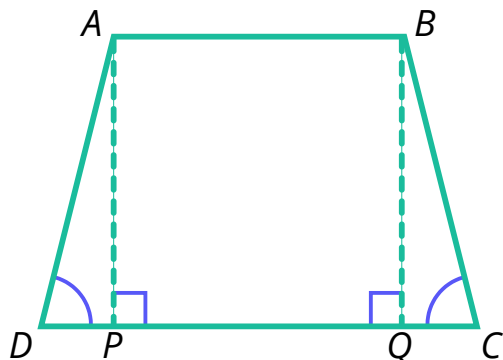
- c Given that $ABCD$ is a parallelogram, and $\overline{BD}$ bisects $\angle ABC$ and $\angle ADC$.



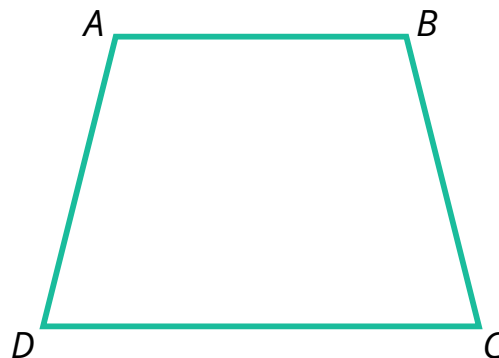
13 Prove that the following is an isosceles trap with the given conditions:



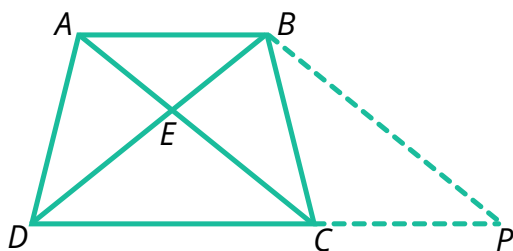
- a Given that $ABCD$ is a trapezoid with altitudes $\overline{AP}$ and $\overline{BQ}$ and that $\angle ADC \cong \angle BCD$.



- b Given that $ABCD$ is a trapezoid such that $\angle A$ and $\angle C$ are supplementary.



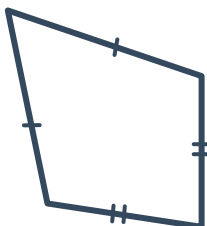
- c Given that $ABCD$ is a trapezoid such that $\overline{AC} \cong \overline{BD}$, and $\overline{AC} \parallel \overline{BP}$.



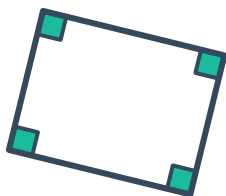
Additional questions

- 14 Classify the following quadrilaterals using all the terms that apply out of: rectangle, rhombus, kite, parallelogram, trapezoid, or neither.

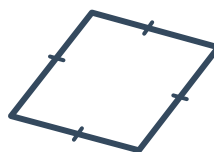
a



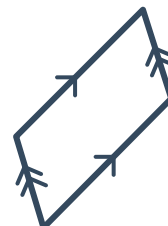
b



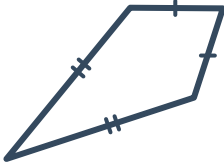
c



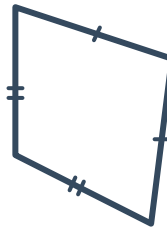
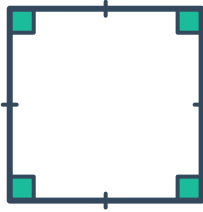
d



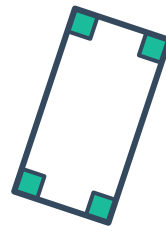
e



f



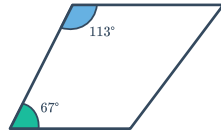
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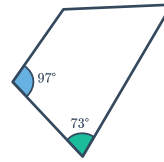
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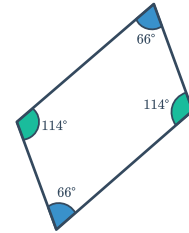
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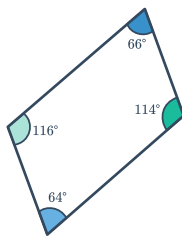
k



l



m



- 15 What is the largest number of obtuse angles that a quadrilateral can have? Explain your answer.
- 16 What is the largest number of right angles that a quadrilateral can have? Explain your answer.
- 17 Can a quadrilateral have exactly 2 right angles? If so, name a quadrilateral that could have two right-angles.
- 18 Can a quadrilateral have exactly 3 right angles?